REP-030 · Rules re-analysis for a tabletop-ready market game (proposal, no rule change)

Game	Fruit Market, rules version 3 (`main` at 416f381)
Report date	2026-10-27 12:59 (Asia/Kuala_Lumpur)
Type	Analysis, lab experiments and a proposal. **Game code unchanged**
Owner request	Re-analyse fruit pricing (lorry and customer), decay, events, traits, power-ups and supply/demand control, against three guidelines: (1) playable on a physical table, (2) a real market feel with inventory, forecasting, strategy and undercutting, balanced all round, (3) each fruit's playstyle in line with its payout
Proposed plan	PLAN-014 "Tabletop edition rules" (awaiting approval)

Labels. [Code] = how the game works now. **[Sim]** = computer-player result: 600 games per test on seeds never used for tuning, about ±4 points. **[Hyp]** = a guess about people, for the playtest.

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1. Verdict against the three guidelines (rules version 3)

Guideline	Today	Main gap
1. Playable on a table	**Partly**	Deliveries and order sizes come from formulas (player count × game progress, rounded); 4-player ×1.15; up to ~10 secret prices per player per day; sliding every fruit one step each night is slow by hand
2. Market feel	**Partly**	Undercutting and forecasting work; prices never react to shortage or glut; fruit is sold within half a day (85–90% bought fresh, 0.3 spoiled per shop per game), so inventory and decay barely matter; shared-loss events (Slow Season, Rainy Day) are luck with no decision
3. Fruit playstyle = payout	**No**	Blueberries earn the best return on cost of all four ($1.72 → $5.61) though they are meant to be the cheap volume fruit; apples (safest) earn more than watermelons

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2. How the rules work now [Code]

2a. Fruit: lorry price, customer price, decay

Fruit	Farmers-market cost fresh / okay / not so / green	Customer top price (24 cards, after shift)	Price drop by age, oldest → freshest	Life on the shelf
Blueberry	2 / 1 / 1 / –	5–8 (shift 0)	−4 · −2 · 0	3 nights
Watermelon	3 / 2 / 1 / 1	6–9 (0)	−3 · −1 · −1 · 0 · 0	5 nights (+1 green)

Fruit	Farmers-market cost fresh / okay / not so / green	Customer top price (24 cards, after shift)	Price drop by age, oldest → freshest	Life on the shelf
Apple	3 / 2 / 1 / –	6–10 (−1)	−2 · −1 · −1 · −1 · 0 · 0 · 0	7 nights
Mango	4 / 2 / 1 / 3	7–11 (+2)	−3 · −1 · 0 · 0	4 nights (+2 green)

- The farmers market cost is fixed per freshness stage; events change it ±1.
- The customer's price is the card value plus the fruit shift plus the event, minus the age drop.
- Fruit at the farmers market ages too, and every night all fruit moves one step.

2b. Supply and demand control (the tabletop problem)

- **Deliveries:** one farmer card a day lists three fruits (4–6 each); the fourth fruit comes as 2. Each number is then multiplied by a formula: 4 per shop × (0.8 + 0.2 × game progress) ÷ 17, rounded. Tomorrow's card is shown a day early.
- **Customers:** one expected customer, known a day early, plus 1 walk-in (2 from halfway), drawn face down. Every order size is multiplied by a formula: 4 per shop × (0.6 + 0.45 × progress), × 1.15 with 4 players, ÷ cards, rounded.
- **Selling:** every shop sets a secret price and quantity for every customer and fruit. Then:
  - cheapest sells first;
  - ties go to a Haggler facing a richer shop, then fresher fruit, then Favoured Trade, then turn order.

2c. What events, traits and upgrades do to outcomes

Events (from REP-025, income vs a normal day, 2p / 4p):

- Big gains: Festival +11 / +8, Superfood +11 / +5, Mango Craze +10 / +4 (the largest gap between shops).
- Neutral: Heatwave, Minimum Price.
- Losses: Slow Season **−12** / −7, Rainy Day −8 / −5, Lorry Breakdown −6 / −3.
- Most events hit every shop equally. They reward planning only when they are announced a day early, which they are.

Traits [Sim, rules 3]:

- 3 players: 30–37%, a 7-point spread (fair 33%).
- 4 players: 23–29%.

Close enough.

Upgrades [Sim]: building only one upgrade, against 10.8% for building none:

Upgrade	Win %
Signboard	20.7
Cool Room	15.8
Handcart	14.7

Upgrade	Win %
Shop Bell	13.5
Juice Press	12.8
Bigger Shelf	12.2

All are worth something; the Signboard is the clear best.

Favour cards: small one-off boosts. Fine on a table.

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3. Proposed "Tabletop edition" (rules version 4)

3a. Core changes (tested)

#	Change	Replaces	Physical form	What decision it improves
T1	**Printed deliveries.** Each farmer card shows 3-player amounts: its three fruits at 2–4 (today's number −2), the fourth fruit at 1. With 2 players, one less of each listed fruit (minimum 1); with 4 players, one more of every fruit	The supply formula	Numbers on the cards plus one line in the rules	None directly; makes the game runnable by hand with almost the same flow
T2	**Printed orders and a walk-in rule.** Customer cards are used as printed. Walk-ins: players − 2 in the first half of the game, players − 1 in the second half	The demand formula and ×1.15	The deck and one line in the rules	Same as T1
T3	**One posted price per fruit.** Each shop secretly sets one price per fruit on a 4-slider price board behind its screen, and puts the fruit it will sell in its "for sale" crate. On reveal, customers (waiting ones first) buy from the cheapest shop whose fruit is fresh enough for the price	~10 secret prices per customer and fruit	A price board and a screen	The core duel becomes "what will they post for mangoes today?", which reads like a real market stall and is quick on a table
T4	**Market pulse.** Each fruit has a marker on a track from −$1 to +$2. After sales: if any customer still wants that fruit, the marker goes up $1 (shortage). If at least players + 1 ripe ones are left at the farmers market, it goes down $1 (glut). Otherwise it drifts one step back to 0. Customers pay the marker on top of their card price	Fixed prices	One cube per fruit on a small track	Prices respond to supply and demand. Spotting tomorrow's shortage pays, and flooding a fruit is punished
F1	**Fruit payouts lined up with risk.** Blueberry customers pay $1 less; apple customers $1 less again (apple shift −2)	Card shifts	Printed on the customer cards	See 3c

3b. Results [Sim, 3 players unless noted]

Measure	Rules 3 now	T1+T2	+ T3 + T4	+ F1 (proposed)
Undercut	27	22	29	28

Measure	Rules 3 now	T1+T2	+ T3 + T4	+ F1 (proposed)
Dump old stock	36	32	32	28
Premium prices	28	30	31	33
Reserve	36	38	33	34
Ripen	34	35	39	38
Read rivals	37	35	38	35
Turnover	23	24	24	26
Events	29	30	24	28
Spread	14	16	15	**12**
4 players: Undercut / Salvage / Premium	20 / 28 / 20	20 / 30 / 22	21 / 28 / 21	23 / 29 / 20
Hard beats Normal (2p)	60	59	63	62
Comeback from last at half-time (3p / 4p)	25.8 / 19.2	28 / 19	32.3 / 19.2	30.2 / 18.0
Price-war sales (3p / 4p)	5.8 / 6.7	5.0 / 7.1	3.9 / 4.2	5.9 / 7.0
Average cash (2p / 3p / 4p)	150 / 108 / 100	137 / 116 / 94	146 / 133 / 116	135 / 122 / 103
4-plan table (Turnover / Reserve / Ripen / Events)	16/32/33/20	16/40/26/19	18/29/35/18	15/29/38/19

Daily flow, formula against printed numbers (fruit delivered / units customers want, first half → second half):

Players	Formula (now)	Printed numbers
3	9.7 → 11.0 / 8.4 → 12.1	10.1 → 10.1 / 8.7 → 13.1
4	13.7 → 15.3 / 12.8 → 17.3	14.1 → 14.1 / 13.1 → 17.4
2	6.8 → 8.0 / 5.5 → 8.1	7.1 → 7.1 / 4.4 → 8.8

The printed rules reproduce the market closely.

Traits under the proposal: 31–37% in 3 players.

Caution. The computers' posted price is a rough stand-in: they take their lowest planned price for each fruit. People will price more cleverly, so T3 must be playtested.

3c. Fruit identities and payout [Sim, proposal]

Fruit	Target playstyle	Cost (fresh)	Average sale	Net per fruit bought	Risk
🥭 Mango	**Speculator:** buy green for $3, wait 2 nights, sell big on Mango Craze or shortage days	$3.61	$9.15	**$5.15**	Highest: 2-night wait, spoils fast once ripe, few delivered

Fruit	Target playstyle	Cost (fresh)	Average sale	Net per fruit bought	Risk
🫐 Blueberry	**Quick flipper:** cheap and plentiful; sell today or lose $2 a night	$1.72	$5.58	**$3.51**	High: 3-night life, steepest decay
🍉 Watermelon	**Steady bulk seller:** big orders, slow decay, a 1-night green bet	$2.51	$6.20	**$3.15**	Medium
🍎 Apple	**The keeper:** lasts a week, sells any day, the safe stock for a Lorry Breakdown or a shortage	$2.79	$6.08	**$3.00**	Lowest

Now the more a fruit can hurt you, the more it pays: mango > blueberry > watermelon > apple. Under rules 3 apples paid more than watermelons, and blueberries gave the best return on cost.

3d. Supporting proposals (not simulated: they need people, or computer players that plan ahead)

1. **The queue: forecasting the crowd.**
    - Tomorrow's walk-ins are drawn tonight and placed face down.
    - The card backs show which fruits each customer wants, but not how many or at what price.
    - With the face-up expected customer and tomorrow's lorry, players can forecast demand: "two walk-ins want apples tomorrow, and the lorry brings few".
    - Physically: printed card backs. Digitally: a row of silhouettes with fruit icons.
2. **Market Days.** Every third day is a busy fair day: +2 walk-ins, and the lorry doesn't come.
    - Shops must stock up the day before, which is where shelf space, decay and apples matter.
    - [Sim] Comebacks rise to 34% and blowouts fall to 31%.
    - But the computers don't plan ahead, and skill matters less in their games (Hard beats Normal 52%).
    - [Hyp] This may be the strongest inventory mechanic, so it should go to a paper playtest before any decision.
3. **Crates instead of sliding.** On a table, fruit goes into a crate stamped with the day it arrived. Its age (today minus that day) gives its freshness from a printed chart. Nothing needs sliding each night, and the digital game already works like this underneath.
4. **Events with decisions.**
    - Keep the events that are announced a day early and reward preparation: Lorry Breakdown, Mango Craze, Superfood, Heatwave, Bad Harvest, Hotel Order.
    - Replace the two pure shared losses (Slow Season, Rainy Day) with single-fruit shocks such as "Blueberry glut: +2 blueberries per player, blueberry customers pay −$1" or "Apple shortage: no apples delivered".
    - Scale Good Harvest by players: +1 of each fruit per 2 players, rounded up.
    - Never draw a night-only event (Inspector, Heatwave ageing) for the final day.
    - Needs a simulation round before numbers are fixed.
5. **Traits and upgrades:** no changes needed for the table; all are printed cards.
    - Juice Press stays the weakest upgrade (12.8%). [Hyp] It gains value with Market Days and stocking ahead.

3e. What it would mean for the app

- Rules version 4: old saves cleared; old apps refused online; separate daily boards (all already handled).
- **Pricing screen:** four fruit price sliders plus "how many to put out", instead of one card per customer. It is simpler and fits the full-screen sheet from PLAN-013.
- **Board:** a market pulse marker on each fruit column, and walk-in silhouettes in the street (if 3d-1 is approved).
- **Computer players:** must learn posted pricing properly before release. Today's stand-in is crude.
- **Rules page and how-to:** rewritten. The digital game and a paper version share one rulebook.

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4. Risks

- **Posted prices remove per-customer pricing.** A shop can no longer charge a rich customer more than a poor one. [Hyp] Simpler and more market-like, but less fine control for experts. Hard beats Normal 62%, still under the 65% aim.
- **Market pulse adds one track and a rule after sales.** [Hyp] It is easy on a table, but it is a new idea for players.
- **The computers can't judge** Market Days, the queue or event redesigns, so those depend on playtests.
- **It is a large change:** a new pricing screen, computer pricing work, and a rules rewrite.

5. Playtest plan

1. **Paper prototype:** print the farmer and customer cards with the new numbers, use coins and tokens, play 3 evenings with 3–4 people. Test T1–T4, the queue and Market Days. Time each day; target 6 minutes or less on a table.
2. **Digital preview:** after building, the same questions as REP-027 section 7, plus:
    - "Did the posted prices feel like a market?"
    - "Did you ever stock up ahead?"

6. Recommendation

Order	Item	Status
1	Approve **PLAN-014 core: T1 + T2 + T3 + T4 + F1** (tested above), with computer posted-price pricing and the new pricing screen	Awaiting approval
2	Paper playtest of the queue, Market Days and crates, then decide	Proposal
3	Event redesign (3d-4) after a simulation round	Proposal

Lab files (engine copy, sweeps 12–15) are archived in `docs/workflow/reports/img/REP-030-lab.tar.gz`.